Mouhamadou Lamine SOW

Computer Science Engineering Student — Machine Learning and Operations Research Internship

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Profile

Final-year engineering student at Sup Galilée, working at the intersection of **machine learning** and **operations research**. Data science experience at **Sonatel (Orange Group)** and projects in computer vision and optimization, built with rigorous evaluation and reproducible code. Available for a **six-month final-year internship from March 2027**.

Professional Experience

Machine Learning / Data Science Intern May 2026 – Jul 2026
Sonatel (Orange Group) Dakar, Senegal

- Designed an XGBoost model predicting network risks 6 months ahead: **88 % fewer false negatives** than the existing business rule (PR-AUC 0.81 vs. 0.28).
- Built, with a teammate, an **8-year fiber network history** (2018–2026); engineered temporal and geographic features with leakage-free chronological evaluation.
- Calibrated probabilities and explained predictions (SHAP); extended to **4 horizons from 3 to 12 months** and mapped risk by area to support network planning.

Job-Shadowing Internship — Digital Technology and DevOps Jul 2024 – Sep 2024
Sonatel (Orange Group) Dakar, Senegal

- Wrote a literature review on **intelligent DevOps automation**; introduction to Angular, LoopBack and MongoDB.

Projects

DermTriage — Dermatological Image Classification 2026
Tools: Python, PyTorch, torchvision, scikit-learn, pytest

- Classified **10,015 images into 7 classes** (HAM10000) under a leakage-free protocol: lesion-grouped splits and a held-out test set.
- Fine-tuned a ResNet-18 network: **macro-F1 of 0.66** vs. 0.26 for a classical baseline (5-fold cross-validation).
- Ran ablation studies to measure each design choice (**+0.15 macro-F1** from deeper fine-tuning); reproducible pipeline, 55 pytest tests.

ROADEF/EURO 2020 — RTE Maintenance Scheduling 2026
Tools: Python, Gurobi, MILP, simulated annealing, local search

- Designed a greedy repair heuristic followed by simulated annealing: **100 % feasible solutions**, verified with the official checker.
- Cut the mean gap to the best known solution to **0.52 %** on 12 test instances (vs. 2.83 % for the greedy baseline).
- Built a Gurobi MILP model strengthened with valid inequalities: **8 proven optima out of 15 instances**.

Education

Engineering Degree in Computer Science — Master's level, expected 2027 Sep 2024 – Present
Sup Galilée — Université Sorbonne Paris Nord Villetaneuse, France
Key coursework: machine learning, deep learning, NLP and Transformers, operations research, software engineering.

Integrated Preparatory Program (CPI) — MPSI Sep 2022 – May 2024
ESIGELEC–ESMT Dakar, Senegal
Mathematics, physics and engineering sciences.

Skills

Programming	Python, SQL, C/C++
Machine Learning	PyTorch, scikit-learn, pandas, NumPy; computer vision, NLP, imbalanced classification
Optimization	MILP, Gurobi, CPLEX, PuLP; metaheuristics, local search
Evaluation	Cross- and time-based validation, calibration, interpretability (SHAP)
Tools	Git, Linux, pytest, Jupyter

Languages & Certification

French: native **English:** B2 — TOEIC 802

IBM AI Engineering Professional Certificate — IBM / Coursera, in progress (2026).